Inverse Image — Prabal Ghosh — Home Work - 2

Problem (I) Consider the following optimization problem,

$$\min_{x \in R} \frac{1}{2}\|Ax-y\|_2^2 + \lambda R_{\alpha,\beta}(x) + i_{\geq 0}(x)$$

when, $R_{\alpha,\beta}(x) = \|x\|_1 + \langle \alpha, x\rangle + \beta, \quad \alpha \in R^n, \beta > 0$

and A is convolution operator $Ax = h * x$

(1) Derive the Proximal operator of the regularization term $R_{\alpha,\beta}$

→ To derive the Proximal operator of the regularization term, $R_{\alpha,\beta}(x) = \|x\|_1 + \langle \alpha, x\rangle + \beta$, we need the definition of Proximal operator.

$$\boxed{\text{prox}_{\gamma R_{\alpha,\beta}}(y) = \underset{x \in R^n}{\text{argmin}} \left\{ \frac{1}{2}\|x-y\|_2^2 + \gamma R_{\alpha,\beta}(x) \right\}}$$

$$= \underset{x \in R^n}{\text{argmin}} \left\{ \frac{1}{2}\|x-y\|_2^2 + \gamma\|x\|_1 + \gamma\langle \alpha, x\rangle + \gamma\beta \right\}$$

The term $\gamma\beta$ is constant with respect to x, so it does not affect the minimizer and can be omitted.

$$\Rightarrow \underset{x \in R^n}{\text{argmin}} \left\{ \frac{1}{2}\|x-y\|_2^2 + \gamma\|x\|_1 + \gamma\langle \alpha, x\rangle \right\}$$

$$= \underset{x \in R^n}{\text{argmin}} \left\{ \frac{1}{2}\|x-y\|_2^2 + \gamma\|x\|_1 + \langle \gamma\alpha, x-y\rangle + \frac{1}{2}\|\gamma\alpha\|_2^2 \right\}$$

$$= \arg\min \left\{ \frac{1}{2} \| x - (y - \gamma\alpha) \|_2^2 + \gamma \|x\|_1 \right\}$$

$$= \text{prox}_{\gamma\|\cdot\|_1} (y - \gamma\alpha)$$

⇒ As, Soft-Thresholding function S_γ:

$$S_\gamma(z)_i = \text{sign}(z_i) \cdot \max(\|z_i\| - \gamma, 0)$$

$$\Rightarrow \boxed{\text{prox}_{\gamma R_{\alpha,\beta}}(y) = S_{\gamma\beta}(y - \gamma\alpha)}$$

→ Shifted soft thresholding function

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② Give the pseudo-code for one iteration of FISTA.

⇒ Objective fun^c ⇒

$$\min_{x \in R^n} \frac{1}{2} \|Ax - y\|_2^2 + \lambda R_{\alpha,\beta}(x) + i_{\geq 0}(x);$$

Algorithm :- FISTA Iteration for Convolution Based Sparse Reconstruction.

Required input ⇒
- Previous iterations: $x_k, x_{k-1} \in R^n$.
- Momentum parameter: $t_k \in R$
- Step size: $\tau \in R^+$
- Regularization parameters: $\lambda, \alpha \in R^n$, $\beta \in R^+$
- Convolution kernel: $h \in R^n$
- Observed data: $y \in R^n$

Prahlad Ghosh

Outputs:-
- Updated iterates: x_{k+1}, t_{k+1}

1. Update Momentum Parameter:- $$t_{k+1} \leftarrow \frac{1 + \sqrt{1 + 4t_k^2}}{2}$$

2. Compute Extrapolated Point y_k:- $$y_k \leftarrow x_k + \left(\frac{t_k - 1}{t_{k+1}}\right)(x_k - x_{k-1})$$

3. Gradient Step in Fourier Domain:-

$$FT(A \cdot y_k) \leftarrow FT(h) \odot FT(y_k)$$

$$FT(r_k) \leftarrow FT(A y_k) - FT(y)$$

$$\nabla f(y_k) \leftarrow IFT\left(FT(h)^* \odot FT(r_k)\right)$$

$$x_{k+1/2} \leftarrow y_k - \gamma \nabla f(y_k)$$

4. Proximal Step for $\lambda R_{\alpha,\beta}(x)$:-

$$x_{k+1}^{temp} \leftarrow S_{\gamma\lambda}\left(x_{k+1/2} - \gamma\lambda\alpha\right)$$

where, $S_\gamma(z)_i = \text{sign}(z_i) \cdot \max(|z_i| - \gamma, 0)$.

5. Project on Non Negative Constraint:-

$$\cancel{x_{k+1} = \max\left(0, S_{\gamma\lambda}\left(x_{k+1/2} - \gamma\lambda\alpha\right)\right)}$$

$$x_{k+1} \leftarrow \max\left(0, x_{k+1}^{temp}\right)$$

4

3. Analyse the role of the parameter β in the reconstruction quality of the solution.

$\Rightarrow$ β does not influence reconstruction quaility; it is an irrelevant additive constant.